2026 03 12
발표 자료

광운대학교 로봇학과
FAIR Lab

김한서

KWANGWOON
UNIVERSITY

이번 주 진행사항

- Implicit Forecaster
  - 실험 세팅 및 결과
  - 아이디어

Implicit Forecaster

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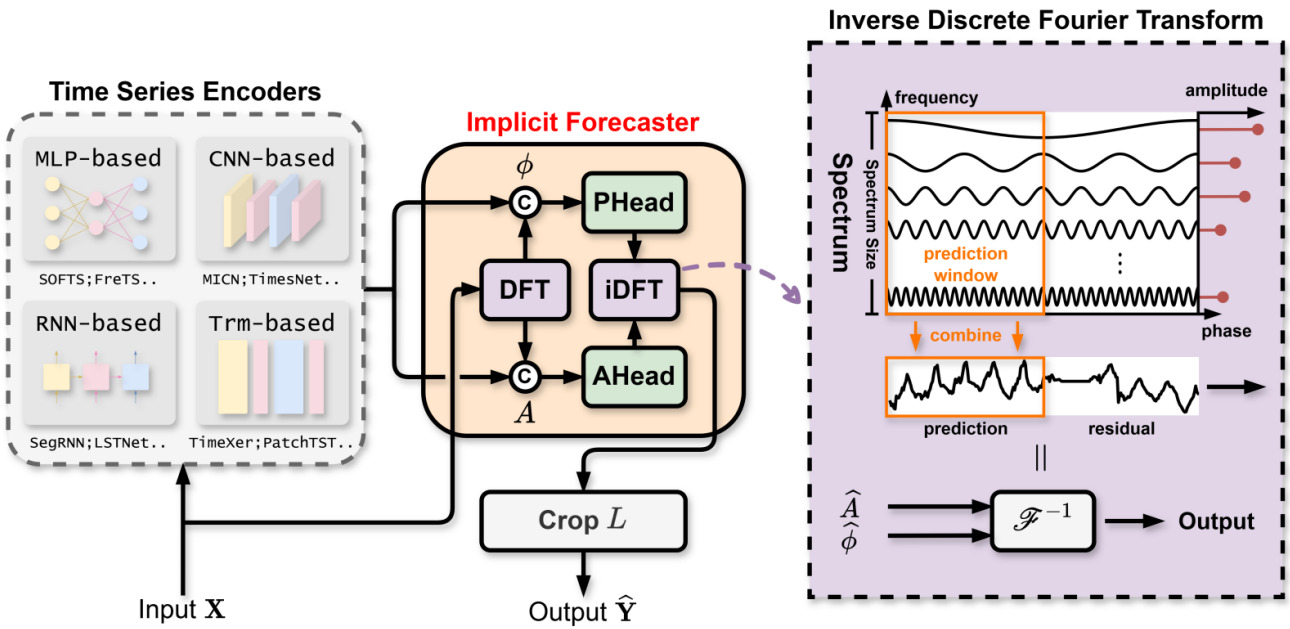
Remind



- 시계열 데이터를 Encoder에 입력하여 변환된 Hidden Representation과 DFT를 통해 추출된 Phase와 Amplitude를 concat하여 사용
- Concat된 데이터를 각 헤드 입력하여 미래 Phase, Amplitude를 예측하고, 각 예측값을 결합한 뒤 iDFT를 통해 시간 도메인으로 복원
- 이후 예측 길이만큼 Crop하여 최종 예측값 생성

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실험 세팅

- 사용한 모델: PatchTST + IF, iTransformer + IF
- 재현 실험 데이터셋: ETTh1

Experiment	ETTh1
Learning rate	10^{-4}
Epoch	100
Batch size	128
Loss function	MSE Loss
Seq_len	336
Pred_len	96/192/336/720
Layers	3
d_model	128
d_ff	512
Patience	10

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PatchTST + IF 재현 실험 (ETTh1)

	ETTh1 Paper		ETTh1 Reproduction	
Pred len	MSE	MAE	MSE	MAE
96	0.411	0.453	0.411	0.454
192	0.485	0.500	0.454	0.480
336	0.497	0.511	0.499	0.511
720	0.601	0.582	0.633	0.591

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iTransformer + IF 재현 실험 (ETTh1)

	ETTh1 Paper		ETTh1 Reproduction	
Pred len	MSE	MAE	MSE	MAE
96	0.456	0.474	0.447	0.462
192	0.490	0.503	0.493	0.495
336	0.533	0.532	0.534	0.520
720	0.660	0.608	0.662	0.595

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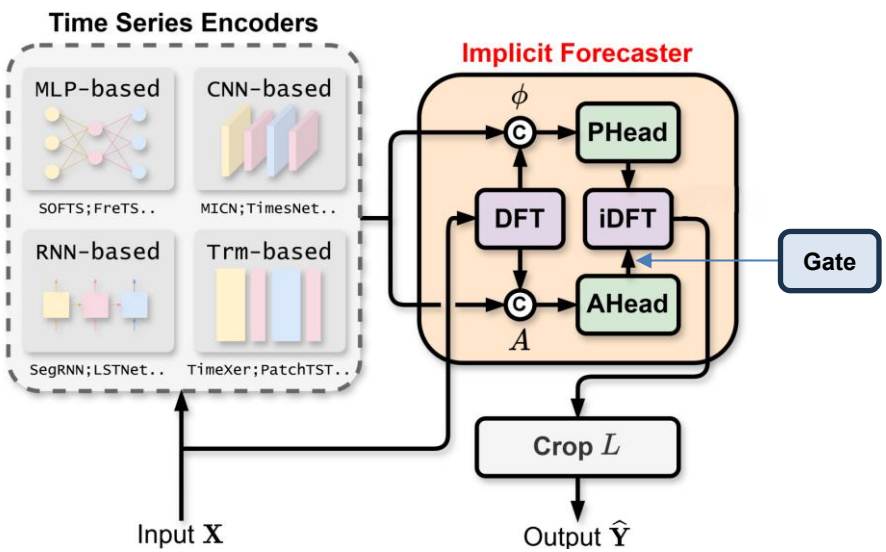

- Ahead의 예측 값을 Learnable Gate Head에 입력하여 진폭이 실제 신호인지 노이즈인지 학습 $g = \sigma\left(MLP_{gate}(\hat{A}_{raw})\right)$, $g \in [0,1]^{N \times P}$
- Sigmoid 함수를 통해 가중치 g로 변환, 기존 Ahead 예측 값과 g를 곱하여 최종 Amplitude 생성 $\hat{A}_{final} = \hat{A}_{raw} \odot g$
- $\mathcal{L}_{amp} = \mathrm{MSE}(\hat{A}_{final}, A_{GT})$, where $A_{GT} = |\mathcal{F}(Y)|$

부록

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주요 모델 성능 비교

Models		SOFTS (2024)		w/ IF (Ours)		TimeXer (2024)		w/ IF (Ours)		iTransformer (2024)		w/ IF (Ours)		SegRNN (2023)		w/ IF (Ours)		PatchTST (2023)		w/ IF (Ours)		Crossformer (2023)		w/ IF (Ours)		TimesNet (2023)		w/ IF (Ours)	
Metric		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	96	0.442	0.461	**0.426**	**0.458**	0.423	**0.457**	**0.420**	0.461	**0.447**	**0.469**	0.456	0.474	**0.432**	**0.463**	0.436	0.467	0.425	0.457	**0.411**	**0.453**	**0.437**	**0.476**	0.437	**0.474**	0.484	0.501	**0.433**	**0.475**
	192	0.493	0.495	**0.469**	**0.489**	0.487	0.503	**0.461**	**0.494**	0.495	**0.503**	**0.490**	**0.503**	0.480	0.497	**0.467**	**0.490**	**0.477**	**0.494**	0.485	0.500	**0.487**	**0.505**	0.581	0.563	0.542	0.536	**0.498**	**0.521**
	336	0.529	**0.519**	**0.510**	0.523	0.529	**0.528**	**0.524**	0.541	0.537	**0.526**	**0.533**	0.532	0.524	0.526	**0.502**	**0.517**	0.552	0.530	**0.497**	**0.511**	**0.640**	0.602	0.644	**0.590**	0.606	0.576	**0.548**	**0.553**
	720	0.667	0.602	**0.618**	**0.595**	0.668	**0.612**	**0.637**	0.618	0.662	**0.601**	**0.660**	0.608	**0.659**	**0.608**	0.689	0.631	0.678	0.607	**0.601**	**0.582**	0.862	0.696	**0.684**	**0.630**	0.752	0.657	**0.662**	**0.637**
	Avg	0.533	0.519	**0.506**	**0.516**	0.527	**0.525**	**0.510**	0.528	**0.535**	**0.525**	**0.535**	0.529	0.524	**0.523**	**0.523**	0.526	0.533	0.522	**0.498**	**0.511**	0.607	0.570	**0.587**	**0.564**	0.596	0.568	**0.535**	**0.546**
ETTh2	96	0.159	0.279	**0.154**	**0.273**	0.153	0.273	**0.151**	**0.270**	0.160	0.280	**0.159**	**0.277**	0.149	**0.269**	**0.148**	0.270	0.152	0.272	**0.147**	**0.266**	0.179	0.310	**0.171**	**0.302**	0.168	0.290	**0.161**	**0.279**
	192	0.188	0.304	**0.182**	**0.295**	0.184	0.301	**0.177**	**0.292**	0.187	0.303	**0.186**	**0.300**	0.180	0.296	**0.175**	**0.295**	0.183	0.302	**0.179**	**0.295**	0.261	0.381	**0.223**	**0.353**	0.209	0.323	**0.199**	**0.309**
	336	0.208	0.320	**0.195**	**0.309**	0.215	0.327	**0.209**	**0.322**	0.208	0.322	**0.202**	**0.315**	0.200	0.315	**0.192**	**0.312**	0.198	0.316	**0.194**	**0.309**	0.256	0.384	**0.251**	**0.371**	0.216	0.332	**0.210**	**0.322**
	720	0.254	0.356	**0.240**	**0.342**	0.263	0.367	**0.239**	**0.341**	0.253	0.356	**0.244**	**0.346**	**0.242**	**0.350**	0.243	0.356	0.234	0.343	**0.227**	**0.333**	0.487	0.505	**0.306**	**0.413**	0.257	0.361	**0.246**	**0.349**
	Avg	0.202	0.315	**0.193**	**0.305**	0.204	0.317	**0.194**	**0.306**	0.202	0.315	**0.198**	**0.309**	0.193	**0.307**	**0.189**	0.308	0.192	0.308	**0.187**	**0.301**	0.296	0.395	**0.238**	**0.360**	0.212	0.327	**0.204**	**0.315**
ETTm1	96	**0.398**	**0.416**	0.406	0.425	**0.417**	**0.432**	0.435	0.438	**0.392**	0.426	0.396	**0.425**	0.400	0.424	**0.389**	**0.416**	**0.390**	0.418	0.403	**0.417**	0.497	0.488	**0.487**	**0.467**	0.487	0.469	**0.463**	**0.460**
	192	**0.438**	**0.448**	0.441	0.454	0.458	0.469	**0.446**	**0.458**	0.447	**0.458**	**0.443**	**0.458**	0.438	0.455	**0.435**	**0.450**	**0.445**	0.455	**0.445**	**0.452**	0.515	0.497	**0.479**	**0.480**	0.514	0.490	**0.478**	**0.476**
	336	0.520	0.492	**0.488**	**0.485**	0.509	0.503	**0.495**	**0.495**	0.523	0.498	**0.499**	**0.493**	**0.492**	0.489	0.495	**0.488**	0.504	0.490	**0.490**	**0.483**	0.541	0.524	**0.531**	**0.517**	0.573	0.534	**0.517**	**0.505**
	720	0.589	0.540	**0.555**	**0.537**	0.577	0.550	**0.534**	**0.529**	0.585	0.540	**0.572**	**0.538**	**0.548**	**0.529**	0.562	0.536	0.571	0.540	**0.544**	**0.527**	**0.600**	**0.554**	0.633	0.580	0.620	0.557	**0.572**	**0.537**
	Avg	0.486	**0.474**	**0.473**	0.475	0.490	0.488	**0.477**	**0.480**	0.487	0.481	**0.478**	**0.478**	**0.470**	0.474	**0.470**	**0.473**	0.477	0.476	**0.471**	**0.470**	0.538	0.516	**0.532**	**0.511**	0.548	0.512	**0.508**	**0.497**
ETTm2	96	0.109	0.227	**0.101**	**0.220**	**0.098**	**0.217**	0.099	**0.217**	0.107	0.227	**0.105**	**0.225**	**0.097**	**0.216**	0.100	0.217	0.102	0.222	**0.100**	**0.219**	0.112	0.240	**0.105**	**0.227**	0.106	0.225	**0.101**	**0.218**
	192	0.130	0.252	**0.128**	**0.249**	0.125	0.247	**0.124**	**0.244**	0.132	0.255	**0.130**	**0.253**	**0.123**	**0.244**	0.125	0.246	0.128	0.251	**0.127**	**0.250**	0.147	0.267	**0.137**	**0.262**	0.136	0.254	**0.130**	**0.248**
	336	0.162	0.280	**0.158**	**0.279**	**0.155**	**0.273**	0.156	**0.273**	0.168	0.286	**0.161**	**0.280**	**0.151**	**0.270**	0.154	0.274	0.158	0.279	**0.152**	**0.272**	0.238	0.344	**0.209**	**0.338**	0.165	0.281	**0.158**	**0.276**
	720	0.204	0.315	**0.202**	**0.314**	0.201	0.311	**0.200**	**0.309**	0.209	0.319	**0.205**	**0.315**	**0.195**	**0.306**	0.198	0.313	0.199	0.310	**0.195**	**0.308**	0.315	0.425	**0.282**	**0.391**	0.215	0.322	**0.203**	**0.313**
	Avg	0.151	0.269	**0.147**	**0.266**	**0.145**	0.262	**0.145**	**0.261**	0.154	0.272	**0.150**	**0.268**	**0.142**	**0.259**	0.144	0.263	0.147	0.266	**0.144**	**0.262**	0.203	0.319	**0.183**	**0.304**	0.155	0.271	**0.148**	**0.264**
ECL	96	0.146	**0.233**	**0.141**	**0.233**	0.142	0.242	**0.137**	**0.231**	0.150	**0.239**	**0.145**	**0.239**	**0.157**	**0.249**	0.165	0.254	0.179	0.271	**0.163**	**0.250**	0.164	0.266	**0.161**	**0.263**	0.167	0.268	**0.160**	**0.265**
	192	**0.157**	**0.245**	**0.157**	0.250	0.155	0.252	**0.151**	**0.247**	0.160	**0.248**	**0.158**	0.253	**0.169**	**0.260**	0.174	0.263	0.184	0.275	**0.170**	**0.259**	0.186	0.284	**0.176**	**0.277**	0.182	0.282	**0.178**	**0.281**
	336	0.170	**0.259**	**0.166**	0.263	0.171	0.271	**0.165**	**0.261**	0.172	**0.263**	**0.169**	0.268	**0.181**	**0.277**	0.185	0.278	0.197	0.289	**0.182**	**0.274**	0.213	0.308	**0.200**	**0.303**	**0.192**	**0.293**	0.194	0.298
	720	0.208	**0.294**	**0.197**	0.296	0.198	0.295	**0.190**	**0.290**	0.208	**0.295**	**0.196**	**0.295**	**0.216**	**0.312**	0.219	**0.312**	0.234	0.321	**0.221**	**0.314**	0.285	0.376	**0.236**	**0.335**	**0.218**	**0.315**	**0.218**	0.320
	Avg	0.170	**0.258**	**0.165**	0.261	0.166	0.265	**0.161**	**0.257**	0.172	**0.261**	**0.167**	0.264	**0.181**	**0.275**	0.186	0.277	0.199	0.289	**0.184**	**0.274**	0.212	0.308	**0.193**	**0.294**	0.190	**0.289**	**0.188**	0.291
Traffic	96	0.393	**0.267**	**0.390**	**0.267**	**0.438**	**0.278**	0.690	0.341	**0.401**	**0.277**	0.432	0.302	0.609	0.313	**0.530**	**0.298**	0.463	0.304	**0.439**	**0.293**	0.544	**0.311**	**0.512**	0.319	0.579	**0.322**	**0.573**	0.348
	192	0.426	0.281	**0.420**	**0.280**	**0.471**	**0.296**	0.664	0.347	**0.427**	**0.288**	0.458	0.311	0.626	0.319	**0.559**	**0.306**	0.478	0.311	**0.460**	**0.302**	0.565	**0.320**	**0.525**	0.329	0.608	**0.332**	**0.584**	0.363
	336	0.444	**0.289**	**0.440**	0.291	**0.487**	**0.298**	0.621	0.347	**0.441**	**0.294**	0.482	0.324	0.644	0.326	**0.571**	**0.316**	0.493	0.317	**0.475**	**0.308**	0.579	**0.328**	**0.548**	0.337	0.622	**0.338**	**0.592**	0.350
	720	0.468	**0.307**	**0.467**	0.309	**0.523**	**0.316**	0.541	0.338	**0.467**	**0.310**	0.617	0.396	0.675	0.345	**0.633**	**0.339**	0.520	0.335	**0.507**	**0.331**	0.610	**0.346**	**0.598**	0.358	0.648	**0.351**	**0.623**	0.376
	Avg	0.433	**0.286**	**0.429**	0.287	**0.480**	**0.297**	0.629	0.343	**0.434**	**0.292**	0.497	0.333	0.639	0.326	**0.573**	**0.315**	0.489	0.317	**0.470**	**0.308**	0.575	**0.326**	**0.546**	0.336	0.614	**0.336**	**0.593**	0.359
Weather	96	0.164	**0.205**	**0.161**	0.214	0.147	**0.194**	**0.145**	0.196	0.164	0.202	**0.154**	**0.201**	**0.156**	**0.217**	0.160	0.218	0.164	**0.205**	**0.158**	0.206	**0.163**	**0.230**	0.164	0.233	0.165	0.213	**0.156**	**0.210**
	192	**0.204**	**0.240**	0.207	0.257	0.191	**0.234**	**0.189**	0.238	0.207	**0.240**	**0.198**	0.242	**0.198**	0.258	0.199	**0.255**	0.207	**0.242**	**0.196**	**0.242**	**0.222**	0.295	0.224	**0.292**	0.207	**0.249**	**0.203**	0.250
	336	0.257	0.278	**0.241**	**0.277**	**0.244**	**0.273**	0.247	0.282	0.259	0.280	**0.244**	**0.277**	0.251	0.300	**0.245**	**0.294**	0.258	0.279	**0.249**	**0.278**	**0.256**	**0.320**	0.263	0.323	0.272	0.293	**0.255**	**0.289**
	720	0.319	**0.322**	**0.309**	0.331	0.304	**0.314**	**0.303**	0.324	0.319	**0.322**	**0.305**	0.323	0.317	0.348	**0.305**	**0.340**	0.320	**0.320**	**0.312**	0.321	**0.327**	**0.377**	0.359	0.404	0.322	0.327	**0.309**	**0.326**
	Avg	0.236	**0.261**	**0.229**	0.270	**0.221**	**0.254**	**0.221**	0.260	0.237	**0.261**	**0.225**	**0.261**	0.230	0.281	**0.227**	**0.277**	0.237	**0.262**	**0.229**	**0.262**	**0.242**	**0.305**	0.253	0.313	0.241	0.270	**0.231**	**0.269**
Exchange	96	0.089	0.214	**0.082**	**0.204**	0.086	0.207	**0.082**	**0.204**	0.103	0.231	**0.102**	**0.229**	**0.082**	**0.200**	0.085	0.208	0.091	0.212	**0.086**	**0.206**	**0.236**	**0.352**	0.262	0.372	0.096	0.225	**0.090**	**0.217**
	192	0.200	0.328	**0.177**	**0.305**	0.187	0.312	**0.177**	**0.306**	0.206	0.334	**0.188**	**0.318**	**0.174**	**0.298**	0.207	0.333	0.184	0.310	**0.183**	**0.309**	**0.427**	**0.484**	0.526	0.539	**0.185**	**0.319**	0.187	0.321
	336	0.389	0.465	**0.349**	**0.437**	**0.371**	**0.449**	0.373	0.452	**0.362**	**0.448**	0.364	0.450	0.360	0.437	**0.306**	**0.424**	0.361	0.445	**0.345**	**0.434**	**1.063**	**0.807**	2.276	1.191	**0.381**	**0.456**	0.382	0.460
	720	1.017	0.809	**0.999**	**0.807**	**0.943**	**0.763**	0.950	0.773	1.038	0.820	**0.994**	**0.798**	1.142	0.848	**0.578**	**0.608**	1.128	0.847	**1.074**	**0.834**	**1.466**	**0.973**	2.287	1.198	1.037	0.825	**1.023**	**0.816**
	Avg	0.424	0.454	**0.402**	**0.438**	0.397	**0.433**	**0.395**	0.434	0.427	0.458	**0.412**	**0.449**	0.440	0.446	**0.294**	**0.393**	0.441	0.454	**0.422**	**0.446**	**0.798**	**0.654**	1.338	0.825	0.425	0.456	**0.420**	**0.454**
ILI	96	2.731	**1.001**	**2.621**	1.035	**2.903**	1.134	2.941	**1.124**	2.820	1.085	**2.543**	**1.014**	3.974	1.272	**2.920**	**1.078**	2.585	1.005	**2.292**	**1.003**	**3.922**	**1.375**	4.598	1.508	**2.715**	**1.054**	2.811	1.057
	192	**2.665**	**1.105**	2.837	1.121	3.052	1.153	**2.677**	**1.119**	3.034	1.173	**2.880**	**1.113**	4.366	1.379	**3.167**	**1.173**	4.506	1.292	**4.131**	**1.252**	**3.971**	**1.383**	5.034	1.573	**3.670**	**1.245**	5.866	1.333
	336	3.485	1.280	**3.212**	**1.212**	3.402	1.235	**2.926**	**1.154**	**3.158**	**1.224**	3.263	1.232	4.425	1.427	**3.932**	**1.339**	**3.578**	1.219	3.736	**1.202**	**3.958**	**1.420**	4.861	1.543	3.633	1.229	**3.537**	**1.228**
	720	2.691	**1.101**	**2.637**	1.111	**3.179**	**1.174**	3.208	1.191	2.953	1.175	**2.797**	**1.116**	5.759	1.651	**4.547**	**1.395**	2.792	1.102	**2.581**	**1.050**	**5.132**	**1.582**	6.021	1.745	3.037	1.120	**2.723**	**1.089**
	Avg	2.893	1.122	**2.827**	**1.120**	3.134	1.174	**2.938**	**1.147**	2.991	1.164	**2.871**	**1.119**	4.631	1.432	**3.641**	**1.246**	3.365	1.155	**3.185**	**1.127**	**4.246**	**1.440**	5.128	1.592	**3.264**	**1.162**	3.734	1.177
PEMS03	12	0.064	0.167	**0.061**	**0.164**	0.064	0.177	**0.058**	**0.165**	0.065	0.172	**0.062**	**0.169**	0.078	0.191	**0.074**	**0.185**	0.092	0.208	**0.070**	**0.178**	0.083	0.194	**0.073**	**0.185**	0.087	0.195	**0.069**	**0.173**
	24	0.081	**0.189**	**0.080**	**0.189**	0.086	0.203	**0.072**	**0.181**	0.093	0.206	**0.085**	**0.197**	0.123	0.243	**0.113**	**0.229**	0.144	0.266	**0.105**	**0.218**	0.117	0.237	**0.104**	**0.224**	0.118	0.222	**0.094**	**0.192**
	48	**0.118**	**0.230**	0.127	0.238	0.130	0.248	**0.110**	**0.220**	**0.160**	**0.273**	0.279	0.360	0.225	0.333	**0.189**	**0.301**	0.275	0.372	**0.177**	**0.285**	0.207	0.321	**0.174**	**0.287**	0.186	0.272	**0.119**	**0.220**
	96	**0.165**	**0.272**	0.167	0.275	0.207	0.311	**0.167**	**0.269**	1.337	0.918	**0.410**	**0.447**	0.356	0.425	**0.261**	**0.364**	0.496	0.518	**0.285**	**0.370**	0.260	0.363	**0.234**	**0.340**	0.309	0.352	**0.142**	**0.248**
	Avg	**0.107**	**0.214**	0.109	0.216	0.122	0.235	**0.102**	**0.209**	0.414	0.392	**0.209**	**0.293**	0.196	0.298	**0.159**	**0.270**	0.252	0.341	**0.159**	**0.263**	0.167	0.279	**0.146**	**0.259**	0.175	0.260	**0.106**	**0.208**
PEMS04	12	**0.080**	**0.185**	0.085	0.193	0.079	0.192	**0.072**	**0.175**	0.094	0.199	**0.084**	**0.189**	0.103	0.215	**0.097**	**0.208**	0.123	0.242	**0.096**	**0.205**	0.095	0.210	**0.088**	**0.200**	0.135	0.251	**0.082**	**0.190**
	24	0.102	0.213	**0.093**	**0.199**	0.095	0.212	**0.085**	**0.193**	0.130	0.236	**0.102**	**0.211**	0.157	0.273	**0.144**	**0.258**	0.196	0.309	**0.139**	**0.252**	0.126	0.248	**0.111**	**0.230**	0.180	0.290	**0.095**	**0.208**
	48	0.125	0.238	**0.116**	**0.227**	0.120	0.241	**0.100**	**0.211**	0.197	0.296	**0.134**	**0.247**	0.265	0.363	**0.225**	**0.330**	0.371	0.430	**0.217**	**0.321**	0.176	0.300	**0.143**	**0.264**	0.240	0.342	**0.125**	**0.243**
	96	0.141	0.254	**0.136**	**0.247**	0.157	0.279	**0.136**	**0.250**	0.328	0.398	**0.178**	**0.288**	0.406	0.462	**0.298**	**0.390**	0.647	0.591	**0.294**	**0.384**	0.529	0.567	**0.184**	**0.301**	0.335	0.418	**0.166**	**0.288**
	Avg	0.112	0.223	**0.107**	**0.216**	0.113	0.231	**0.098**	**0.207**	0.187	0.282	**0.124**	**0.234**	0.233	0.328	**0.191**	**0.296**	0.334	0.393	**0.186**	**0.290**	0.232	0.331	**0.132**	**0.249**	0.222	0.325	**0.117**	**0.232**
PEMS07	12	0.067	**0.163**	**0.062**	**0.163**	0.064	0.169	**0.055**	**0.150**	0.067	0.167	**0.060**	**0.158**	0.079	0.187	**0.073**	**0.176**	0.089	0.209	**0.072**	**0.176**	0.097	0.204	**0.076**	**0.188**	0.105	0.222	**0.068**	**0.169**
	24	0.083	0.185	**0.074**	**0.174**	0.084	0.193	**0.069**	**0.170**	0.098	0.201	**0.081**	**0.185**	0.130	0.244	**0.117**	**0.227**	0.152	0.270	**0.116**	**0.225**	0.141	0.252	**0.113**	**0.227**	0.130	0.248	**0.090**	**0.191**
	48	0.103	**0.204**	**0.102**	0.207	0.108	0.212	**0.080**	**0.181**	1.511	1.012	**0.205**	**0.309**	0.241	0.338	**0.206**	**0.306**	0.303	0.385	**0.201**	**0.297**	0.255	0.331	**0.178**	**0.282**	0.179	0.294	**0.116**	**0.227**
	96	**0.130**	**0.230**	0.131	0.237	0.142	0.247	**0.107**	**0.210**	1.860	1.138	**0.250**	**0.346**	0.393	0.439	**0.300**	**0.380**	0.556	0.539	**0.302**	**0.370**	0.291	0.374	**0.255**	**0.339**	0.253	0.355	**0.161**	**0.273**
	Avg	0.096	**0.195**	**0.092**	**0.195**	0.100	0.205	**0.078**	**0.178**	0.884	0.629	**0.149**	**0.249**	0.211	0.302	**0.174**	**0.272**	0.275	0.351	**0.173**	**0.267**	0.196	0.290	**0.155**	**0.259**	0.167	0.280	**0.109**	**0.215**
PEMS08	12	0.076	0.179	**0.070**	**0.173**	0.093	0.211	**0.074**	**0.184**	0.087	0.193	**0.078**	**0.184**	0.093	0.203	**0.085**	**0.192**	0.102	0.218	**0.082**	**0.188**	0.098	0.205	**0.085**	**0.195**	0.125	0.240	**0.081**	**0.189**
	24	0.107	0.211	**0.098**	**0.207**	0.134	0.256	**0.101**	**0.217**	0.132	0.242	**0.110**	**0.220**	0.145	0.258	**0.127**	**0.240**	0.168	0.279	**0.122**	**0.231**	0.145	0.260	**0.116**	**0.231**	0.161	0.277	**0.098**	**0.210**
	48	0.153	0.248	**0.123**	**0.230**	0.140	0.256	**0.100**	**0.208**	1.649	1.082	**0.228**	**0.335**	0.253	0.351	**0.202**	**0.311**	0.321	0.406	**0.199**	**0.305**	0.197	0.311	**0.153**	**0.271**	0.219	0.326	**0.134**	**0.251**
	96	**0.136**	**0.236**	0.137	0.237	0.210	0.318	**0.156**	**0.261**	1.571	0.986	**0.302**	**0.397**	0.437	0.467	**0.290**	**0.385**	0.614	0.581	**0.287**	**0.379**	0.439	0.493	**0.216**	**0.331**	0.335	0.416	**0.199**	**0.318**
	Avg	0.118	0.218	**0.107**	**0.212**	0.144	0.260	**0.108**	**0.217**	0.860	0.626	**0.179**	**0.284**	0.232	0.320	**0.176**	**0.282**	0.301	0.371	**0.172**	**0.276**	0.220	0.317	**0.142**	**0.257**	0.210	0.315	**0.128**	**0.242**
Solar	96	0.192	0.233	**0.185**	**0.231**	0.207	0.268	**0.202**	**0.250**	0.210	0.253	**0.202**	**0.243**	0.226	0.290	**0.218**	**0.273**	0.211	0.271	**0.210**	**0.256**	**0.193**	**0.257**	**0.193**	0.259	0.211	**0.236**	**0.198**	0.253
	192	0.223	**0.253**	**0.215**	0.255	0.241	0.293	**0.240**	**0.276**	0.240	0.277	**0.225**	**0.268**	0.244	0.304	**0.240**	**0.288**	0.247	0.294	**0.235**	**0.272**	0.231	**0.281**	**0.224**	0.287	0.256	0.288	**0.221**	**0.274**
	336	0.238	0.263	**0.218**	**0.262**	0.257	0.303	**0.243**	**0.282**	0.258	0.291	**0.246**	**0.287**	0.259	0.304	**0.248**	**0.291**	0.257	0.296	**0.237**	**0.281**	0.249	0.298	**0.236**	**0.293**	0.288	0.282	**0.222**	**0.280**
	720	0.234	0.264	**0.210**	**0.263**	0.251	0.297	**0.237**	**0.281**	0.255	0.291	**0.238**	**0.285**	0.249	0.295	**0.245**	**0.291**	0.246	0.293	**0.233**	**0.278**	0.522	0.539	**0.235**	**0.294**	0.285	0.295	**0.218**	**0.271**
	Avg	0.222	**0.253**	**0.207**	**0.253**	0.239	0.290	**0.231**	**0.272**	0.241	0.278	**0.228**	**0.271**	0.244	0.298	**0.238**	**0.286**	0.240	0.288	**0.229**	**0.272**	0.299	0.344	**0.222**	**0.283**	0.260	0.273	**0.212**	**0.270**